Figure 8.13. A high-level overview of the intrinsic curiosity module (ICM). The ICM has three components that are each separate neural networks. The encoder model encodes states into a low-dimensional vector, and it is trained indirectly through the inverse model, which tries to predict the action that was taken given two consecutive states. The forward model predicts the next encoded state, and its error is the prediction error that is used as the intrinsic reward.

Figure 8.14. This figure shows the type and dimensionality of the inputs and outputs of each component of the ICM.

Figure 8.15. The DQN and the ICM contribute to a single overall loss function that is given to the optimizer to minimize with respect to the DQN and ICM parameters. The DQN's Q-value predictions are compared to the observed rewards. The observed rewards, however, are summed together with the ICM's prediction error to get a new reward value.

Figure 8.16. A complete view of the overall algorithm, including the ICM. First we generate B samples from the experience replay memory and use these for the ICM and DQN. We run the ICM forward to generate a prediction error, which is then provided to the DQN's error function. The DQN learns to predict action values that reflect not only extrinsic (environment provided) rewards but also an intrinsic (prediction error-based) reward.

Figure 8.17. These are the losses for the individual components of the ICM and the DQN. The losses do not smoothly decrease like we're used to with a single supervised neural network because the DQN and ICM are trained adversarially.

Figure 8.18. The DQN agent and forward model are trying to optimize antagonistic objectives and hence form an adversarial pair.

Figure 8.19. Training time is on the x axis and episode length is on the y axis. We see bigger and bigger spikes over training time, which is what we expect.

Figure 8.20. The Mario agent trained only from intrinsic rewards successfully jumping over a chasm. This demonstrates it has learned basic skills without any explicit rewards to do so. With a random